Exact big Ramsey degrees for arbitrary ordinals below omega to the omega

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Editorial note. Original complete argument retained with version/license/figure dependencies; all support belongs to this single counted target.

Editorial note. Prepared omega-power result has a single uniform level. This target handles mixed exponents and finite coefficients with generalized rules and a new block construction; intermediate omega^d times k claims are not separately counted.

Allowed standard prerequisites (editorial).

- Standard subject prerequisites. Standard named subject results cited by the authors may be used; all paper-specific supporting arguments are retained verbatim. No modified assumptions or replacement proof.

1 Introduction

Definition 1.1. Let $\mathbb{N}$ be the set of natural numbers, including 0. Let $n, c \in \mathbb{N}$ and S be a set. Then we define $[c]$ to be $\{0, \ldots, c-1\}$ (note $0 \in [c]$, but $c \notin [c]$). Note that if $c = 0$ then $[c] = \emptyset$. We use $\binom{S}{n}$ to denote the set of all n-element subsets of S.

Definition 1.2. A function $\text{COL}\colon \binom{S}{n} \to [c]$ is called a *coloring*. A set $H \subseteq S$ is called *homogeneous* if COL restricted to $\binom{H}{n}$ is constant, that is, $\left|\text{COL}\left(\binom{H}{n}\right)\right| \leq 1$.

Ramsey's Theorem on infinite countable sets [27] is as follows:

Theorem 1.3. *Let $n, c \in \mathbb{N}$, and let S be an infinite countable set. For all colorings $\text{COL}\colon \binom{S}{n} \to [c]$ there exists an infinite homogeneous set $H \subseteq S$.*

If S is an *ordered* set, then Ramsey's Theorem does not guarantee a homogeneous set of the same order type. Instead, we might be able to get a set $H \subseteq S$ of the same order type where $\left|\text{COL}\left(\binom{H}{n}\right)\right|$ is bounded by some constant. Mašulović and Šobot [22] prove the following:

Theorem 1.4. *Let $n, c \in \mathbb{N}$, and let S be a set of order type ζ (the integers). For all colorings $\text{COL}\colon \binom{S}{n} \to [c]$ there exists an infinite homogeneous set $H \subseteq S$ such that $\left|\text{COL}\left(\binom{H}{n}\right)\right| \leq 2^n$. Note that 2^n is independent of c.*

Our main results are about the colorings of finite subsets of countable ordinals less than ω^ω. We provide necessary definitions in Section 2, and summarize our results rigorously in Section 3.

2 Definitions

Definition 2.1. Let $\mathcal{A} = (A, \preceq_A)$ and $\mathcal{B} = (B, \preceq_B)$ be ordered sets, where $\preceq_A, \preceq_B$ are total orders on A, B respectively.

1. Two ordered sets $\mathcal{A}, \mathcal{B}$ are *order-equivalent*, denoted $\mathcal{A} \approx \mathcal{B}$, if there exists an order-preserving bijection $f\colon A \to B$ such that, for all $a_1, a_2 \in A$:

$$a_1 \preceq_A a_2 \iff f(a_1) \preceq_B f(a_2).$$

2. The ordered set $\mathcal{A} + \mathcal{B}$ is defined to be $(A \sqcup B, \preceq)$, where the disjoint union $A \sqcup B$ is formally defined to be $(\{0\} \times A) \cup (\{1\} \times B)$, and where $\preceq$ is the dictionary ordering on $A \sqcup B$. Note that $+$ agrees with the definition of ordinal addition, and is not commutative in general.

3. Let $\mathcal{A} = (A, \preceq_A)$ be an ordered set with $A \subseteq \mathbb{Z}$. Then define $-\mathcal{A}$ to be $(-A, \preceq_{-A})$, where $-A = \{-a\colon a \in A\}$, and $-a \preceq_{-A} -b \iff b \preceq_A a$.

Throughout this paper, we conflate the notation for an ordered set and its underlying set. For example, if A is an ordered set, we might still use the notation $\binom{A}{2}$.

Definition 2.2. Let S be an ordered set, $S' \subseteq S$, and $n, c, t \in \mathbb{N}$. Let $\mathrm{COL}\colon \binom{S}{n} \to [c]$ be a coloring.

1. S' is *homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| = 1$.

2. S' is *t-homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| \leq t$.

3. S' is *S-t-homogeneous* if both S' is t-homogeneous and $S' \approx S$.

Definition 2.3. We identify every ordinal α as the ordered set of all ordinals $\beta < \alpha$, with 0 as the least ordinal. Additionally, let ω be the ordered set containing only the naturals, ζ be the ordered set containing only the integers, and η be the ordered set containing only the rationals, all under their respective natural orderings.

Definition 2.4. Let S be an ordered set. For any $n \in \mathbb{N}$, $T(n, S)$ is the least $t \in \mathbb{N}$ such that for all constants $c \geq 1$ and colorings $\mathrm{COL}\colon \binom{S}{n} \to [c]$, there exists some $S' \subseteq S$ such that S' is S-t-homogeneous. Note that t is independent of c. If no such t exists then we define $T(n, S) = \infty$. The quantity $T(n, S)$ is called the *big Ramsey degree* of n in S. The term was first coined by Kechris et al. [17]. Other sources use the Ramsey arrow notation modified for ordered sets. In that notation, $T(n, S) = t$ is short-hand for both $S \to (S)^n_{c,t}$ and $S \not\to (S)^n_{T(n,S),t-1}$ for all $c \in \mathbb{N}$. We use the $T(n, S)$ notation in this paper.

The following stems from Definition 2.4, but we include it for completeness.

Lemma 2.5. *For all ordered sets S, $T(0, S) = 1$.*

Ramsey's Theorem on $\mathbb{N}$ gives an infinite 1-homogeneous subset of $\mathbb{N}$. Theorem 2.6 restates Ramsey's Theorem using big Ramsey degree notation.

Theorem 2.6. *For all $n \in \mathbb{N}$, $T(n, \omega) = 1$.*

3 Related Work and our Work

We summarize what is known about the big Ramsey degrees of linear orders, and what our contribution is. We also briefly discuss big Ramsey degrees for other structures. We then give a more detailed account of our contributions. All of the structures in this section are countable.

3.1 Big Ramsey Degrees of Linear Ordering and Other Structures

The table below summarizes what is known about the big Ramsey degrees of ordinals. We use β to denote an arbitrary ordinal and n for an arbitrary positive number.

Ordinal	Known Results	Comments
ω	$T(n, \omega) = 1$ [27]	Infinite Ramsey Theorem
ω^β	$T(1, \omega^\beta) = 1$ [15] (page 189)	
β	$T(1, \beta) < \infty$ [15] (page 189)	
$\beta < \omega^\omega$	$T(n, \beta) < \infty$ [22]	We provide exact numbers
$\beta \geq \omega^\omega$	$(\forall n \geq 2)[T(n, \beta) = \infty]$ [22]	

Definition 3.1. Let L be a countable scattered linear ordering. Then $\mathrm{HR}(L)$ denotes the Hausdorff rank of L.

In the table below, the ith Bernoulli number is denoted by B_i. We use L to denote an arbitrary countable scattered linear ordering.

Ordering	Known Results	Comments
ζ	$T(n, \zeta) = 2^n$ [22]	We provide an easier proof
η	$T(n, \eta) = \frac{B_{2n+1}(-1)^{n+1}(1-4^{n+1})}{(2(n+1))!}$ [8, 29]	
L	$T(1, L) < \infty$ [20]	
L	$(\forall n \geq 2)[\mathrm{HR}(L) = \infty \iff T(n, L) = \infty]$ [3, 21]	

Big Ramsey degrees have also been studied for structures more complex than linear orders:

1. the generic partial order [1, 16],
2. the Rado graph [14, 18, 19, 26, 28],
3. the Rado hypergraph [2, 5, 6, 7], and
4. the Henson graph [9, 10, 12].

More abstract versions of these questions have also been studied [23, 25, 30]. For a survey and more references, see Dobrinen [11] and Coulson et al. [6, 7]. Page 52 of [7] has a table of known results. For a collection of open problems, see Dobrinen and Gasarch [13].

Exact values are known for coloring edges of the Rado graph: upper bounds by Pouzet & Sauer [26] and lower bounds by Erdős, Hajnal and Posa [14]. When coloring arbitrary sets of n vertices in the Rado graph, exact values are known: see Leflamme, Sauer, and Vuksanovic [18] and Larson [19]. For all $a \geq 3$, the Rado a-hypergraph has finite big Ramsey degree; however, no exact numbers are known. For all k, the k-Henson graph has finite big Ramsey degree. The exact big Ramsey degrees of 3-Henson graphs are known, see Dobrinen [10, Corollary 5.13]. Dobrinen has made a conjecture about the Ramsey degree of k-Henson graphs [12, Section 11: Future Directions], and also supplies some numerical results.

3.2 Summary of Results

Our work is most similar to that of Mašulović & Šobot [22], who study the conditions under which $T(n, \beta)$ is finite. Their methods rely on what they call a top-down approach, where they prove that the property of finite big Ramsey degrees is passed from ordinals of the form ω^n to their subsums of lesser ordinals. Our methods are bottom-up: we provide explicit formulas for calculating the degree of an ordinal number from its subsums. This enables us to compute exact results, unlike the work of Mašulović & Šobot.

In other words, ordinal numbers less than ω^ω can be viewed as summations of smaller and simpler ordinals of the form ω^n. Our main result relies on the big Ramsey degrees of these simpler ordinals to determine the big Ramsey degrees of all ordinals less than ω^ω.

In this paper, we do the following.

1. We show that $T(n, \zeta) = 2^n$ in Section 4. This can be obtained by the result due to Mašulović & Šobot [22] that $T(n, \omega + \omega) = 2^n$, but we give a direct proof using different methods.

2. We find the big Ramsey degrees $T(n, \omega \cdot k)$ for finite k in Section 5.

3. We find the big Ramsey degree $T(2, \omega^2)$ in Section 6 and introduce our main method for finding big Ramsey degrees of ordinals in Section 7.

4. We find the big Ramsey degrees $T(n, \omega^d)$ and $T(n, \omega^d \cdot k)$ for finite d, k in Sections 8 and 9.

5. We find the big Ramsey degrees of all ordinals $\alpha < \omega^\omega$ in Section 10.

4 Big Ramsey Degrees of ζ

Our general result is $T(n, \zeta) = 2^n$. We first prove $T(1, \zeta) = 2$ and $T(2, \zeta) = 4$. Both of these will build notation and structure toward the main result.

Theorem 4.1. *The big Ramsey degree $T(1, \zeta) = 2$.*

Proof. We first prove $T(1, \zeta) \leq 2$. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \zeta \to [c]$$

be a c-coloring of ζ. Let $\mathrm{COL}'\colon \omega \to [c] \times [c]$ be defined by

$$\mathrm{COL}'(x) = (\mathrm{COL}(-x), \mathrm{COL}(x)).$$

By Theorem 2.6 (Ramsey's theorem), there exists a set $G \subseteq \omega$ that is ω-1-homogeneous with respect to COL'. Let the color of the homogeneous set be (c_1, c_2). Consider the set $H = -G + G$, which is order-equivalent to ζ. Let $h \in H$. If $h \in G$ then by the definition of COL' and 1-homogeneity of G, $\mathrm{COL}(h) = c_2$. Otherwise, $h \in -G$, and so $\mathrm{COL}(h) = c_1$. Thus H is ζ-2-homogeneous. Because COL was arbitrary, $T(1, \zeta) \leq 2$.

We now prove $T(1, \zeta) \geq 2$. Let $\mathrm{COL}\colon \zeta \to [2]$ be the function that maps all nonnegative integers to 0 and all negative integers to 1. Since the nonnegative integers have no infinitely descending chain and the negative integers have no infinitely ascending chain, there is no ζ-1-homogeneous subset of ζ under COL. Therefore $T(1, \zeta) \geq 2$, and with the previous result, $T(1, \zeta) = 2$. □

Theorem 4.2. *The big Ramsey degree $T(2, \zeta) = 4$.*

Proof. We first prove $T(2, \zeta) \leq 4$. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\zeta}{2} \to [c]$$

be a c-coloring of $\binom{\zeta}{2}$. Let $\mathrm{COL}'\colon \binom{\omega}{2} \to [c]^4$ be defined by

$$\mathrm{COL}'(x, y) = (\mathrm{COL}(-x, -y), \mathrm{COL}(-x, y), \mathrm{COL}(x, -y), \mathrm{COL}(x, y))$$

where $x < y$.

By Theorem 2.6 there exists an ω-1-homogeneous set G. Let the color of the homogeneous set be (c_1, c_2, c_3, c_4). Let $G = \{h_0 < h_1 < \cdots\}$.

Then consider the set

$$H = \{-h_i\colon i \text{ is even}\} + \{h_i\colon i \text{ is odd}\},$$

which is order-equivalent to ζ. Let $s_i n_i, s_j n_j \in H$, where $n_i, n_j \geq 0$ and $s_i, s_j \in \{-1, 1\}$. Then by definition of COL', $\mathrm{COL}(s_i n_i, s_j n_j)$ is in $\{c_1, c_2, c_3, c_4\}$, depending on s_i and s_j. As an aside, this method only works when n_i and n_j are guaranteed to be distinct, which we forced with our alternation of sign by the parity of index. Since $\mathrm{COL}(H) \subseteq \{c_1, c_2, c_3, c_4\}$, H is ζ-4-homogeneous. Therefore $T(2, \zeta) \leq 4$.

We now prove $T(2,\zeta) \geq 4$. Let $\mathrm{COL}\colon \binom{\zeta}{2} \to [4]$ be the coloring

$$\mathrm{COL}(x,y) = \begin{cases} 0 & x,y \geq 0 \\ 1 & x \geq 0,\ y < 0, \text{ and } |x| \leq |y| \\ 2 & x \geq 0,\ y < 0, \text{ and } |x| > |y| \\ 3 & x < 0,\ y < 0. \end{cases}$$

Let H be an arbitrary order-equivalent subset of ζ. The following x, y are guaranteed to exist because $H \subseteq \zeta$ and $H \approx \zeta$:

1. Let $x, y \in H$ such that $x, y \geq 0$. Then $\mathrm{COL}(x,y) = 0$.
2. Let $x, y \in H$ such that $x \geq 0$, $y < 0$, and $|x| \leq |y|$. Then $\mathrm{COL}(x,y) = 1$.
3. Let $x, y \in H$ such that $x \geq 0$, $y < 0$, and $|x| > |y|$. Then $\mathrm{COL}(x,y) = 2$.
4. Let $x, y \in H$ such that $x < 0$, $y < 0$. Then $\mathrm{COL}(x,y) = 3$.

Therefore $T(2,\zeta) \geq 4$, and with the previous result, $T(2,\zeta) = 4$. □

The proof of Theorem 4.2 exposes the common structure of all the proofs in this paper. We identify $T(n,\alpha)$ ways to reshape some naturals into specific formats, and define a COL' using those cases. After applying Ramsey's theorem to yield some order-equivalent subset of naturals G, we rebuild some order-equivalent H whose coloring under COL is guaranteed by the $T(n,\alpha)$ cases. Our upper bound proofs use the same cases again.

Theorem 4.3. *For all $n \in \mathbb{N}$, $T(n,\zeta) = 2^n$.*

Proof. We first prove $T(n,\zeta) \leq 2^n$. Let $c \in \mathbb{N}$ and let $\mathrm{COL}\colon \binom{\zeta}{n} \to [c]$ be a c-coloring of $\binom{\zeta}{n}$. Let $\mathrm{COL}'\colon \binom{\omega}{n} \to [c]^{(2^n)}$ be defined by

$$\mathrm{COL}'(x_1,\ldots,x_n) = (\mathrm{COL}(x_1,\ldots,x_n), \mathrm{COL}(-x_1,x_2,\ldots,x_n), \ldots, \mathrm{COL}(-x_1,\ldots,-x_n))$$

where $x_1 < x_2 < \cdots < x_n$. Formally, COL' outputs a tuple with COL's output on each item in the set

$$\{-x_1,x_1\} \times \{-x_2,x_2\} \times \cdots \times \{-x_n,x_n\}.$$

Note that COL' only depends on the color of elements of $\binom{\zeta}{n}$ where the absolute values of the integers are all different.

By Theorem 2.6 there exists some homogeneous set G. Let $G = \{h_0 < h_1 < \cdots\}$. Then the set

$$H = \{-h_i\colon i \text{ is even}\} + \{h_i\colon i \text{ is odd}\}$$

is ζ-2^n-homogeneous. Because the absolute values of every element in H are all different, each subset with size n was considered by COL'. Therefore $T(n,\zeta) \leq 2^n$.

We now prove $T(n,\zeta) \geq 2^n$. We describe a coloring $\mathrm{COL}\colon \binom{\zeta}{n} \to [2]^n$.

Let $\{a_1 < \cdots < a_n\} \in \binom{\zeta}{n}$. Define an ordering $<^*$ as $x <^* y$ if $|x| < |y|$ or both $|x| = |y|$ and $x < y$ (we order by absolute values, and in case of ties, order the negative before the positive).

Let $(i_1,\ldots,i_n)$ be such that

$$a_{i_1} <^* a_{i_2} <^* \cdots <^* a_{i_n}.$$

Let s_{i_j} be 0 if $a_{i_j} < 0$ and 1 if $a_{i_j} \geq 0$. We define $\mathrm{COL}(\{a_1 < \cdots < a_n\})$ as $(s_{i_1},\ldots,s_{i_n})$. We use 2^n colors, as there are n elements colored either 0 or 1 in the output of COL. We leave it to the reader to show that there is no ζ-(2^n-1)-homogeneous set.

□

5 Big Ramsey Degrees of Finite Multiples of ω

As noted in Theorem 2.6, $T(n,\omega)=1$. In this and later sections we examine limit ordinals larger than ω. For simplicity in stating results, we save the big Ramsey degrees of successor ordinals such as $\omega+1,\omega+2,\ldots$ as well as some limit ordinals such as $\omega^2+\omega$ for Section 10.

Our first result shows that $T(n,\omega\cdot k)=k^n$ for all n,k where at least one is nonzero. Lemma 2.5 shows that $T(0,\omega\cdot 0)=T(0,\emptyset)=1$.

Theorem 5.1. *For $n,k\in\mathbb{N}$ with at least one of n,k nonzero, $T(n,\omega\cdot k)\leq k^n$.*

Proof. Let $n,c,k\in\mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\omega\cdot k}{n}\to[c]$$

be a c-coloring of $\binom{\omega\cdot k}{n}$. Let $\mathrm{COL}'\colon \binom{\omega}{n}\to[c]^{(k^n)}$ be defined by

$$\begin{aligned}\mathrm{COL}'(x_1,x_2,\ldots,x_n)=(&\mathrm{COL}(x_1,\ldots,x_n),\mathrm{COL}(\omega+x_1,x_2,\ldots,x_n),\ldots,\\&\mathrm{COL}(\omega\cdot(k-1)+x_1,\ldots,\omega\cdot(k-1)+x_n))\end{aligned}$$

with $x_1<x_2<\cdots<x_n$. Our constructed coloring COL' maps n elements of ω to the COL of each of the k^n ways to add one of $\omega\cdot 0$ through $\omega\cdot(k-1)$ with each of the n coordinates. Formally, its output tuple contains COL's assignment of each element of

$$\{x_1,\omega+x_1,\ldots,\omega\cdot(k-1)+x_1\}\times\cdots\times\{x_n,\omega+x_n,\ldots,\omega\cdot(k-1)+x_n\}.$$

Apply Theorem 2.6 with COL' to find some $G\approx\omega$ such that

$$\left|\mathrm{COL}'\left(\binom{G}{n}\right)\right|=1.$$

Let the one color in $\mathrm{COL}'\left(\binom{G}{n}\right)$ be Y. Note that Y is a tuple of length k^n.

Enumerate G in order as $\{g_0<g_1<\cdots\}$ and let

$$\begin{aligned}H=&\{\omega\cdot 0+g_i\colon i\equiv 0 \mod k\}+\\&\{\omega\cdot 1+g_i\colon i\equiv 1 \mod k\}+\cdots+\\&\{\omega\cdot(k-1)+g_i\colon i\equiv k-1 \mod k\}.\end{aligned}$$

For the case of $k=3$, we have

$$\begin{aligned}H=\{\quad & g_0, && g_3, && g_6,\ldots,\\ & \omega+g_1, && \omega+g_4, && \omega+g_7,\ldots,\\ & \omega\cdot 2+g_2, && \omega\cdot 2+g_5, && \omega\cdot 2+g_8,\ldots\}.\end{aligned}$$

Now $H\approx\omega\cdot k$. Then

$$\left|\mathrm{COL}\left(\binom{H}{n}\right)\right|\leq k^n\colon$$

for any selection of n elements from H, its color was considered in COL' so it must be one of the k^n colors in Y. □

We now prove that $T(n,\omega\cdot k)$ is bounded below by k^n by providing an example coloring. The coloring is inspired by Theorem 5.1. This duality will be made more clear in later sections.

Theorem 5.2. *For $n,k\in\mathbb{N}$, $T(n,\omega\cdot k)\geq k^n$. Therefore, by Theorem 5.1, $T(n,\omega\cdot k)=k^n$.*

Proof. We give a k^n-coloring of $\binom{\omega \cdot k}{n}$ that has no (k^n-1)-homogeneous $H \approx \omega \cdot k$. We represent $\omega \cdot k$ as

$$\omega \cdot k \approx A_1 + \cdots + A_k$$

where each $A_i \approx \omega$. If an element $x \in \omega \cdot k$ is the kth element of A_i, we represent it as the ordered pair (i,k).

Before defining the coloring in general, we give an example with $n = 5$ and $k = 200$. We define the color of the element

$$e = \{(3,12),(50,2),(110,12),(110,7777),(117,3)\}$$

as follows:

- Order the ordered pairs by their second coordinates.
- If some elements have the same second coordinate, break the tie with their first coordinates. At this point in our example we have

$$((50,2),(117,3)(3,12),(110,12),(110,7777)).$$

- Remove the second coordinates to get

$$(50,117,3,110,110).$$

  This sequence is the element's color.

In general, for any $e = \{(i_1,x_1),\ldots,(i_n,x_n)\}$, order the ordered pairs by their second coordinates, and in case of ties, use their first coordinates to determine the ordering. Then $\mathrm{COL}(e)$ is the sequence of first coordinates after ordering.

Note that the number of colors is the number of size n tuples where each number is in $\{1,\ldots,k\}$. Hence there are k^n colors. We leave it to the reader to show that there can be no $(\omega \cdot k)$-(k^n-1)-homogeneous H. The key idea of the proof, much like the previous lower bounds in this paper, is to use the property of order-equivalence to find elements of an arbitrary H that produce each of the k^n colors.

By this proof, $T(n,\omega \cdot k) \geq k^n$, and so by Theorem 5.1, $T(n,\omega \cdot k) = k^n$.

□

6 The Big Ramsey Degree $T(2,\omega^2)$

This section provides a concrete example involving ordinals greater than ω, in particular ω^2, the ordered set of all ordinals $\omega \cdot a + b$ with $a,b \in \omega$. We first define some notation.

Definition 6.1. If $e \in \binom{\omega^d}{n}$, we refer to e as an *edge*. Then e has n *elements* selected from ω^d. Hence each element of e is some ordinal number less than ω^d. That element can be represented as a polynomial of degree $< d$ in ω. We refer to the natural-valued coefficients of these polynomials as the *coefficients* of e.

Proving $T(1,\omega^2) = 1$ is a straightforward exercise. We begin with $T(2,\omega^2)$ before proceeding to the general case.

Theorem 6.2. *The big Ramsey degree $T(2,\omega^2) = 4$.*

Proof. We first prove $T\left(2,\omega^2\right) \leq 4$.

Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\omega^2}{2} \to [c]$$

be a c-coloring of $\binom{\omega^2}{2}$. We define four functions f_0, f_1, f_2, f_3 from domain $\binom{\omega}{4}$ to codomain $\binom{\omega^2}{2}$ and then use them to define a coloring from $\binom{\omega}{4}$ to $[c]^4$. In what follows, let $x_1 < x_2 < x_3 < x_4$. For $i \in [4]$, we define $f_i \colon \binom{\omega}{4} \to \binom{\omega^2}{2}$ where

$$\begin{aligned} f_0(x_1, x_2, x_3, x_4) &= \{\omega \cdot x_1 + x_2, \omega \cdot x_3 + x_4\} \\ f_1(x_1, x_2, x_3, x_4) &= \{\omega \cdot x_1 + x_3, \omega \cdot x_2 + x_4\} \\ f_2(x_1, x_2, x_3, x_4) &= \{\omega \cdot x_1 + x_4, \omega \cdot x_2 + x_3\} \\ f_3(x_1, x_2, x_3, x_4) &= \{\omega \cdot x_1 + x_2, \omega \cdot x_1 + x_3\}. \end{aligned}$$

Then let $\mathrm{COL}' \colon \binom{\omega}{4} \to [c]^4$ where

$$\mathrm{COL}'(X) = (\mathrm{COL}(f_0(X)), \mathrm{COL}(f_1(X)), \mathrm{COL}(f_2(X)), \mathrm{COL}(f_3(X))).$$

Apply Theorem 2.6 on COL' to find some $G \approx \omega$ where $\left|\mathrm{COL}'\left(\binom{G}{4}\right)\right| = 1$. Let $G = \{a_0 < a_1 < \cdots\}$. Let

$$\begin{aligned} H = \quad & \{\omega \cdot a_1 + a_2, \omega \cdot a_1 + a_6,\ \omega \cdot a_1 + a_{10}, \ldots\} \\ + \ & \{\omega \cdot a_3 + a_4, \omega \cdot a_3 + a_{12}, \omega \cdot a_3 + a_{20}, \ldots\} \\ + \ & \{\omega \cdot a_5 + a_8, \omega \cdot a_5 + a_{24}, \omega \cdot a_5 + a_{40}, \ldots\} \\ & \vdots \end{aligned}$$

Formally,

$$H = A_1 + A_2 + \cdots$$

where

$$A_i = \{\omega \cdot a_{2i-1} + a_j \colon j = 2^i + k2^{i+1}, k \in \mathbb{N}\}.$$

Note that each of the A_i's coefficients (as defined in Definition 6.1) are disjoint.

Then $H \approx \omega^2$, as it is the concatenation of countably infinite sets order-equivalent to ω. Note that for every $\omega \cdot a_i + a_j \in H$ we have $a_i < a_j$.

Let the single 4-tuple that COL' outputs on $\binom{G}{4}$ be Y. Consider any edge $\{\omega \cdot b_1 + b_2, \omega \cdot b_3 + b_4\} \in \binom{H}{2}$ with $\omega \cdot b_1 + b_2 < \omega \cdot b_3 + b_4$.

- If $b_1 \neq b_3$, then the two elements are from different A_i sets. Then $b_1 < b_3$ by the ordering of the two elements and $b_2 \neq b_4$ by the construction of H. We also have $b_1 < b_2$, $b_1 < b_4$, and $b_3 < b_4$ by the construction of H. There are three subcases:
  - $b_1 < b_2 < b_3 < b_4$: then $\mathrm{COL}(f_0(b_1, b_2, b_3, b_4)) \in Y$.
  - $b_1 < b_3 < b_2 < b_4$: then $\mathrm{COL}(f_1(b_1, b_3, b_2, b_4)) \in Y$.
  - $b_1 < b_3 < b_4 < b_2$: then $\mathrm{COL}(f_2(b_1, b_3, b_4, b_2)) \in Y$.

  In all subcases, $\mathrm{COL}(\{\omega \cdot b_1 + b_2, \omega \cdot b_3 + b_4\}) \in Y$.
- If $b_1 = b_3$, then the two elements are from the same A_i. Then $b_2 < b_4$ by the ordering of the elements and so $b_1 = b_3 < b_2 < b_4$ by the construction of H. Because $\mathrm{COL}(f_3(b_1, b_2, b_4, b_4 + 1)) \in Y$ (note that the output of f_3 does not depend on its final argument), $\mathrm{COL}(\{\omega \cdot b_1 + b_2, \omega \cdot b_3 + b_4\}) \in Y$.

In all cases, $\mathrm{COL}(\{\omega \cdot b_1 + b_2, \omega \cdot b_3 + b_4\}) \in Y$ so

$$\mathrm{COL}\left(\binom{H}{2}\right) \subseteq Y$$

with $|Y| = 4$. Because $H \approx \omega^2$ and COL was arbitrary, $T\left(2, \omega^2\right) \leq 4$.

We now prove $T\left(2,\omega^2\right) \geq 4$. Let $\text{COL}\colon \binom{\omega^2}{2} \to [4]$ with

$$\text{COL}(\omega \cdot a_1 + a_2, \omega \cdot a_3 + a_4) = \begin{cases} 0 & a_1 < a_2 < a_3 < a_4 \\ 1 & a_1 < a_3 < a_2 < a_4 \\ 2 & a_1 < a_3 < a_4 < a_2 \\ 3 & \text{otherwise} \end{cases}$$

where $\omega \cdot a_1 + a_2 < \omega \cdot a_3 + a_4$. Let $H \subseteq \omega^2$ be any set such that $H \approx \omega^2$. We show that $\left|\text{COL}\left(\binom{H}{2}\right)\right| = 4$. Every element of H is of the form $\omega \cdot b + e$, for finite b, e. Because $H \approx \omega^2$, we have $H = H_1 + H_2 + \cdots$ where each $H_i \approx \omega$. For each H_i, there exists some b_i such that

$$H_i' = \{\omega \cdot b_i + e \in H_i\} \approx \omega.$$

Let $e_{i,j}$ be such that every element of H_i' is of the form $\omega \cdot b_i + e_{i,j}$, where

$$b_1 < b_2 < \cdots$$

and for all i,

$$e_{i,1} < e_{i,2} < \cdots.$$

We now show each color is output by COL on H.

- Color 0: Starting with any b_i, find some j where $e_{i,j} > b_i$. This is guaranteed, as b_i is finite and the $e_{i,j}$ are infinitely increasing. Then find some k where $b_k > e_{i,j}$; again guaranteed because $e_{i,j}$ is finite and the b_i are infinitely increasing. Finally, find an ℓ where $e_{k,\ell} > b_k$ guaranteed by similar means. Then
$$\text{COL}(\omega \cdot b_i + e_{i,j}, \omega \cdot b_k + e_{k,\ell}) = 0.$$
- Colors 1 and 2 are guaranteed to exist by arguments similar to the argument for color 0.
- Color 3: The case of $a_1 = a_3 < a_2 < a_4$ falls into the category of *otherwise*. With this in mind, we can start with some b_i, and then find a $e_{i,j} > b_i$. Then, we only need to find a $e_{i,\ell} > e_{i,j}$; this is guaranteed because $e_{i,j}$ is finite. Then
$$\text{COL}(\omega \cdot b_i + e_{i,j}, \omega \cdot b_i + e_{i,\ell}) = 3.$$

□

7 Coloring Rules

We introduce a concept called *coloring rules* (hereafter CRs) to prove general results about $T(n, \omega^d \cdot k)$. The concept behind CRs is built on the ideas of Blass et al. [4]. We motivate the concept by examining a previous proof.

The proof that $T(2,\omega^2) = 4$ (Theorem 6.2) used four functions f_0, f_1, f_2, and f_3. These functions were chosen to cover H in a way where the color of every edge in H was output by one of f_0, f_1, f_2, or f_3. Note that our proof did *not* use $f\colon \binom{\omega}{4} \to \binom{\omega^2}{2}$ with

$$f(x_1, x_2, x_3, x_4) = \{\omega \cdot x_1 + x_3, \omega \cdot x_2 + x_3\},$$

for $x_1 < x_2 < x_3 < x_4$.

We did not use f in the proof of $T(2,\omega^2) = 4$ because f does not cover any edges in H: we constructed H in a way where distinct copies of ω had distinct finite coefficients. When $x_1 \neq x_2$, no matter the values of x_1, x_2, and x_3, the elements $\omega \cdot x_1 + x_3$ and $\omega \cdot x_2 + x_3$ cannot both be from H. Our goal in lower bound proofs is to use as few of these f_i functions as possible to cover some carefully constructed H.

We define a notion of colorings that each of f_0, f_1, f_2, and f_3 satisfy for $\binom{\omega^2}{2}$. We also show how to count these colorings, and how these colorings are linked to big Ramsey degrees.

Definition 7.1. We now define CRs (coloring rules) rigorously. We impose a structure on edges and list criteria that CRs must satisfy. For $n, d, k \in \mathbb{N}$, an edge $e = \{p_1, \ldots, p_n\}$ in $\binom{\omega^d \cdot k}{n}$ consists of n elements of $\omega^d \cdot k$. Each element p_i is equal to

$$\omega^d \cdot b_i + \omega^{d-1} \cdot a_{i,d-1} + \omega^{d-2} \cdot a_{i,d-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0},$$

where $a_{i,j} \geq 0$ and $0 \leq b_i < k$.

Thus, any edge e is defined by the n values of the b_i and the $n \cdot d$ values of the $a_{i,j}$.

A *CR (coloring rule)* on $\binom{\omega^d \cdot k}{n}$ is a pair $(\overline{B}, \preceq_{\overline{A}})$ of constraints on these values b_i and $a_{i,j}$ satisfying certain properties we enumerate below. Firstly, $\overline{B}$ is an assignment of the values for the b_i; formally, it is a map $\overline{B} : [n] \to \{0, \ldots, k-1\}$ from indices of the b_i to the values of the b_i. Having $\overline{B}(i) = v$ means we constrain $b_i = v$. Meanwhile, $\preceq_{\overline{A}}$ is a total preorder (i.e. every two of elements are comparable, and $\preceq_{\overline{A}}$ is reflexive and transitive) on the indices of the $a_{i,j}$. Having $(i,j) \preceq_{\overline{A}} (i',j')$ means we constrain $a_{i,j} \leq a_{i',j'}$.

We often denote $\overline{B}$ using an ordered list of clauses:

$$b_1 = \overline{B}(1), b_2 = \overline{B}(2), \ldots, b_n = \overline{B}(n).$$

We often denote $\preceq_{\overline{A}}$ by a permutation of the expressions $a_{1,0}, \ldots, a_{n,d-1}$ interspersed by either $<$ or $=$. We write $a_{i,j} < a_{i',j'}$ to mean that both $(i,j) \preceq_{\overline{A}} (i',j')$ and $(i',j') \not\preceq_{\overline{A}} (i,j)$ and we write $a_{i,j} = a_{i',j'}$ to mean both $(i,j) \preceq_{\overline{A}} (i',j')$ and $(i',j') \preceq_{\overline{A}} (i,j)$. Note that $x = y < z$ and $y = x < z$ are two representations of the same preorder, so this notation is not unique. One example of such a representation is

$$a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

We refer to the expressions $a_{i,j}$ and b_i when used in our notation for $\overline{B}$ and $\preceq_{\overline{A}}$ as variables, and will describe different preorders $\preceq_{\overline{A}}$ by discussing permutations of some sequence of $a_{i,j}$.

To be a CR the following must hold:

1. If $d \geq 1$, then for all $i < i'$, $a_{i,0} < a_{i',0}$. Otherwise when $d = 0$, $b_i < b_{i'}$ for all $i < i'$. (The element indices are ordered by their lowest-exponent variable.)
2. For all i, i', $(\exists j \ a_{i,j} = a_{i',j}) \implies b_i = b_{i'}$. (Two elements with any a values that are the same must have the same b value.)
3. For all $j > j'$, $a_{i,j} < a_{i,j'}$. (The high-exponent variables of each element are strictly less than the low-exponent variables.)
4. $a_{i,j} = a_{i',j'} \implies j = j'$. (Only variables with the same exponent can be equal.)
5. $a_{i,j} \neq a_{i',j} \implies a_{i,j-1} \neq a_{i',j-1}$ for all $j > 0$. (Two elements that differ in a high-exponent variable differ in all lower-exponent variables.)

An example of a CR for $\binom{\omega^2}{2}$ is

$$b_1 = 0, \ b_2 = 0, \ a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

Note that because $k = 1$ in the example, we must have $b_i = 0$ for every i.

Definition 7.2. Two CRs are *equivalent* whenever their constraints $\overline{B}, \preceq_{\overline{A}}$ are exactly equal.

Definition 7.3. The *size* of a CR is how many equivalence classes its $a_{i,j}$ form under $\preceq_{\overline{A}}$: for example, $a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}$ would have size $p = 3$ regardless of b_i. Therefore a CR's size p can be no larger than $n \cdot d$.

Definition 7.4.

1. $P_p\left(n, \omega^d \cdot k\right)$ is the number of CRs of size p there are for $\binom{\omega^d \cdot k}{n}$.

2. $P\left(n, \omega^d \cdot k\right)$ is the total number of CRs there are for $\binom{\omega^d \cdot k}{n}$ of any size. It can be calculated as

$$\sum_{p=0}^{n \cdot d} P_p(n, \omega^d \cdot k).$$

Eventually, we will show $T(n, \omega^d \cdot k) = P(n, \omega^d \cdot k)$.

Definition 7.5. An edge

$$\{\omega^d \cdot b_i + \omega^{d-1} \cdot a_{i,d-1} + \omega^{d-2} \cdot a_{i,d-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0} \colon 1 \leq i \leq n\}$$

satisfies some CR if $\overline{B}(i) = b_i$ for every $1 \leq i \leq n$ and if $(i,j) \preceq_{\overline{A}} (i',j') \iff a_{i,j} \leq a_{i',j'}$ for all $1 \leq i, i' \leq n$ and $0 \leq j, j' < d$. Note that while every edge describes some map $\overline{B}$ and order $\preceq_{\overline{A}}$, if those constructs do not meet the requirements in Definition 7.1, that edge does not satisfy any CR. We extend and formalize the idea of coefficients introduced in Definition 6.1 to be the set of $a_{i,j}$ terms.

8 Big Ramsey Degrees of ω^d

This section is devoted to the case where $k = 1$ in $\binom{\omega^d \cdot k}{n}$. When $k = 1$, each b_i in a CR is forced to be 0. Then all b_i values are the same, so criterion 2 of Definition 7.1 is always satisfied. In this section, our proofs focus only on how $\preceq_{\overline{A}}$ permutes the $a_{i,j}$ variables. When values for b_i are not specified, they are assumed to be all 0 by default.

We show equality between big Ramsey degrees and counts of CRs in this section. We start with a recurrence that counts CRs.

Lemma 8.1. *For $n, d \in \mathbb{N}$,*

$$P_p\left(n, \omega^d\right) = \begin{cases} 0 & (1) \quad d = 0 \wedge n \geq 2 \\ 1 & (2) \quad n = 0 \wedge p = 0 \\ 0 & (3) \quad n = 0 \wedge p \geq 1 \\ 1 & (4) \quad d = 0 \wedge n = 1 \wedge p = 0 \\ 0 & (5) \quad d = 0 \wedge n = 1 \wedge p \geq 1 \\ 1 & (6) \quad d = 1 \wedge n \geq 1 \wedge n = p \\ 0 & (7) \quad d = 1 \wedge n \geq 1 \wedge n \neq p \\ 0 & (8) \quad d \geq 2 \wedge n \geq 1 \wedge p = 0 \\ \sum\limits_{j=1}^{n} \sum\limits_{i=0}^{p-1} \binom{p-1}{i} P_i\left(j, \omega^{d-1}\right) P_{p-1-i}\left(n-j, \omega^d\right) & (9) \quad d \geq 2 \wedge n \geq 1 \wedge p \geq 1 \end{cases}$$

Proof. **Case 1.** First, suppose $n \geq 2$ and $d = 0$. As argued at the beginning of this section, $b_i = 0$ for all i. But by criterion 1 of Definition 7.1, since $d = 0$ we need $b_1 < b_2$, so no CRs are possible regardless of size p. This proves the first case of the result.

Case 2. Suppose $n = 0$. Since there are no b_i or $a_{i,j}$, the criteria are vacuously satisfied. There is only one CR and it has size $p = 0$. This proves the second and third cases of the result.

Case 3. When both $d = 0$ and $n \leq 1$, criterion 1 of Definition 7.1 is vacuously satisfied with either no b_i or $b_1 = 0$. Again, because $n \cdot d = 0$, there are no $a_{i,j}$. Thus there is only one CR of size $p = 0$, which proves the fourth and fifth cases of the result.

Case 4. Now suppose $n \geq 1$ and $d = 1$. To ensure criterion 1 of Definition 7.1, each of the n variables $a_{i,0}$ can only form one CR $a_{1,0} < a_{2,0} < \cdots < a_{n,0}$ of size n so $P_n\left(n, \omega^d\right) = 1$ and $P_p\left(n, \omega^d\right) = 0$ for $p \neq n$. This proves the sixth and seventh cases of the result.

Case 5. Finally, consider $n \geq 1, d \geq 2$. By the definition of a CR, because $n \cdot d > 0$, there are no CRs of size $p = 0$. This proves the eighth case of the result. We prove the final case by showing that the process described below creates all possible CRs of an expression.

Let $n \geq 1$, $d \geq 2$, and $p \geq 1$ be naturals. Let $1 \leq m \leq n$ and $0 \leq s \leq p-1$ be naturals. As we proceed through the process, we work with an example of $n = 4$, $d = 5$, $p = 13$, $m = 2$, and $s = 5$.

We create

$$\binom{p-1}{s} P_s\left(m, \omega^{d-1}\right) P_{p-1-s}\left(n-m, \omega^d\right)$$

CRs of size p by combining $P_s\left(m, \omega^{d-1}\right)$ CRs of size s and $P_{p-1-s}\left(n-m, \omega^d\right)$ CRs of size $p-1-s$, with $\binom{p-1}{s}$ CRs derived from each pair.

Let τ_1 represent one of the $P_s\left(m, \omega^{d-1}\right)$ CRs of $\binom{\omega^{d-1}}{m}$ of size s, and τ_2 represent one of the $P_{p-1-s}\left(n-m, \omega^d\right)$ CRs of $\binom{\omega^d}{n-m}$ of size $p-1-s$. In our example, let

$$\begin{aligned}
&\tau_1\colon a_{1,3} = a_{2,3} < a_{1,2} = a_{2,2} < a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0} \\
&\tau_2\colon a_{1,4} = a_{2,4} < a_{1,3} = a_{2,3} < a_{1,2} = a_{2,2} < a_{2,1} < a_{1,1} < a_{1,0} < a_{2,0}.
\end{aligned}$$

Then we can combine each τ_1 and τ_2 to form $\binom{p-1}{s}$ unique new CRs of size p. In summary, we will interleave the rules together, and add a final equivalence class at the start of the CR. Change the index of each variable $a_{i,j}$ in τ_2 to $a_{i+m,j}$, and intertwine the equivalence classes of the CRs together, preserving each CR's original ordering of its own equivalence classes: there are $\binom{p-1}{s}$ ways to do this. In our example, after changing the indices of τ_2 we have

$$\begin{aligned}
&\tau_1\colon a_{1,3} = a_{2,3} < a_{1,2} = a_{2,2} < a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0} \\
&\tau_2\colon a_{3,4} = a_{4,4} < a_{3,3} = a_{4,3} < a_{3,2} = a_{4,2} < a_{4,1} < a_{3,1} < a_{3,0} < a_{4,0}
\end{aligned}$$

and one of the $\binom{12}{5}$ permutations is

$$\begin{aligned}
a_{3,4} = a_{4,4} < a_{1,3} = a_{2,3} < a_{1,2} = a_{2,2} < a_{3,3} = a_{4,3} < a_{3,2} = a_{4,2} \\
< a_{1,1} = a_{2,1} < a_{4,1} < a_{1,0} < a_{3,1} < a_{3,0} < a_{2,0} < a_{4,0}.
\end{aligned}$$

This new CR likely breaks criterion 1 of Definition 7.1; for each $1 \leq i \leq n$, change the i indices of each $a_{i,j}$ according to where their $a_{i,0}$ is in the ordering of all $a_{i,0}$. In our example, we have $a_{1,0} < a_{3,0} < a_{2,0} < a_{4,0}$; after swapping indices 2 and 3 to enforce criterion 1, we have

$$\begin{aligned}
a_{2,4} = a_{4,4} < a_{1,3} = a_{3,3} < a_{1,2} = a_{3,2} < a_{2,3} = a_{4,3} < a_{2,2} = a_{4,2} \\
< a_{1,1} = a_{3,1} < a_{4,1} < a_{1,0} < a_{2,1} < a_{2,0} < a_{3,0} < a_{4,0}.
\end{aligned}$$

There are now $d \cdot (n-m) + (d-1) \cdot m = n \cdot d - m$ variables in the CR. There are m variables of the form $a_{i_s,d-1}$ for $1 \leq s \leq m$ that are not in the CR yet; insert one equivalence class $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$ at the front of the new CR, bringing its size to p. We insert $a_{1,4} = a_{3,4}$ in our example to get

$$\begin{aligned}
a_{1,4} = a_{3,4} < a_{2,4} = a_{4,4} < a_{1,3} = a_{3,3} < a_{1,2} = a_{3,2} < a_{2,3} = a_{4,3} < a_{2,2} = a_{4,2} \\
< a_{1,1} = a_{3,1} < a_{4,1} < a_{1,0} < a_{2,1} < a_{2,0} < a_{3,0} < a_{4,0}.
\end{aligned}$$

Each CR is unique by the τ_1 and τ_2 used to create it because the process is invertible: we can remove the leading equivalence class and separate the remaining variables into τ_1 and τ_2 by whether their indices were in the leading equivalence class and changing the indices. Here, τ_1 corresponds to indices 1 and 3 (not including the leading equivalence class) and is **bolded**, and τ_2 corresponds to indices 2 and 4 and is underlined.

$$\begin{aligned}
a_{1,4} = a_{3,4} < \underline{a_{2,4} = a_{4,4}} < \mathbf{a_{1,3} = a_{3,3}} < \mathbf{a_{1,2} = a_{3,2}} < \underline{a_{2,3} = a_{4,3}} < \underline{a_{2,2} = a_{4,2}} \\
< \mathbf{a_{1,1} = a_{3,1}} < \underline{a_{4,1}} < \mathbf{a_{1,0}} < \underline{a_{2,1}} < \underline{a_{2,0}} < \mathbf{a_{3,0}} < \underline{a_{4,0}}.
\end{aligned}$$

We now show that each CR created by this process has the properties described by Definition 7.1. Because the high-exponent equivalence class $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$ was added at the start of the CR, the high-exponent coefficients of each term are smaller than the low-exponent coefficients (see Definition 7.5 for a definition of coefficients). Criterion 1 is satisfied by changing the indices of the variables. The remaining

criteria are satisfied because τ_1 and τ_2 satisfied them and their internal orders were preserved in permuting the equivalence classes. Therefore this process does not overcount CRs.

We next show that every CR of $\binom{\omega^d}{n}$ is counted by this process. Each can be mapped to some τ_1 and τ_2 that create it by the argument 2 paragraphs above which proves that the process creates unique CRs. Every CR of $\binom{\omega^d}{n}$ must have a leading equivalence class of $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$ to satisfy Definition 7.1 (the equivalence class might only contain one variable); taking only the variables $a_{i,j}$ with indices appearing in that equivalence class (but not those variables in the equivalence class itself) forms τ_1, a CR for $\binom{\omega^{d-1}}{m}$. The variables with i indices not in the equivalence class form τ_2, a CR for $\binom{\omega^d}{n-m}$. The original CR of $\binom{\omega^d}{n}$ is counted by interleaving τ_1 with τ_2 and inserting the leading equivalence class of $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$. Therefore the final case of the result holds. □

For more information, see the OEIS [24], where $P(n, \omega^2)$ is sequence A000311 and $P(2, \omega^d)$ is A079309. Both of these sequences have combinatorial interpretations outside of Ramsey theory. Values for small n, d are tabulated in the appendix; see Table 1. This table has been added to the OEIS, see sequence A364026. It is worth noting that the recurrence

$$\sum_{j=1}^{n} \sum_{i=0}^{p-1} \binom{p-1}{i} P_i\left(j, \omega^{d-1}\right) P_{p-1-i}\left(n-j, \omega^d\right)$$

can be simplified using exponential generating functions of the form

$$F(n, \omega^d)[x] = \sum_{p=0}^{\infty} \frac{1}{p!} P_p(n, \omega^d) x^p.$$

8.1 $T(n, \omega^d) \leq P(n, \omega^d)$

We use the following lemma to show that CRs bound big Ramsey degrees from above.

Lemma 8.2. *For $n, d \in \mathbb{N}$ and some ordered set $G \subseteq \omega$ with $G \approx \omega$, there exists some $H \subseteq \omega^d$ with $H \approx \omega^d$ where for all $e \in \binom{H}{n}$, e satisfies a CR of $\binom{\omega^d}{n}$ and each coefficient of e (see Definition 7.5) is contained in G.*

Proof. Let $G = \{a_1 < a_2 < a_3 < \cdots\}$ with each $a_i \in \omega$ and $G \approx \omega$. We proceed by induction on d.

When $d = 0$, $\omega^0 \approx 1$ and so $H = \{0\} = 1$ suffices. When $n < 2$, the only CR of $\binom{H}{n} = \binom{1}{n}$ is satisfied by every edge of $\binom{1}{n}$ and has $n \cdot d = 0$ coefficients: the 1 in $\omega^0 \cdot 1$ is a b_i term, not an $a_{i,j}$. When $n \geq 2$, there are no CRs of $\binom{1}{n}$ so the result is vacuously satisfied.

When $d = 1$, $\omega^1 \approx \omega$ so $H = G$ suffices, as the single CR for $\binom{\omega}{n}$ is satisfied by all n-subsets of ω, and their n coefficients are in H, hence in G.

For $d \geq 2$, partition G into infinite sets, each order-equivalent to ω:

$$\begin{aligned} A_0 &= \{a_1, a_3, a_5, \ldots\} \\ A_1 &= \{a_2, a_6, a_{10}, \ldots\} \\ A_2 &= \{a_4, a_{12}, a_{20}, \ldots\} \\ A_3 &= \{a_8, a_{24}, a_{40}, \ldots\} \\ &\vdots \end{aligned}$$

Formally,

$$A_i = \{a_j : j = 2^i + k2^{i+1}, k \in \mathbb{N}\}.$$

Reserve A_0 for later. Apply the inductive hypothesis, taking G to be A_i, for all $i \geq 1$. This yields $S_i \approx \omega^{d-1}$ for all $i \geq 1$. Then for all $i \geq 1$, for all $e \in \binom{S_i}{n}$, e satisfies a CR of $\binom{\omega^{d-1}}{n}$ and each coefficient of e is

contained in A_i. We now pair each element of A_0 with an S_i. Let

$$\begin{aligned} H = &\quad \{\omega^{d-1}a_1 + \beta : \beta \in S_0\} \\ &\cup \{\omega^{d-1}a_3 + \beta : \beta \in S_1\} \\ &\cup \{\omega^{d-1}a_5 + \beta : \beta \in S_2\} \\ &\cup \cdots \\ &= \bigcup_{i=0}^{\omega} H_i, \text{ where } H_i = \{\omega^{d-1}a_{2i+1} + \beta : \beta \in S_i\}, \end{aligned}$$

ordered by ordinal comparison. Every element of H_i is less than every element of H_{i+1}. Also, each H_i is order-isomorphic to S_i and thus ω^{d-1}. Therefore, $H \approx \omega^d$.

For any edge e in $\binom{H}{n}$, index its variables to satisfy criterion 1 of Definition 7.1 (this is possible because all low-exponent coefficients are distinct in H). Because we constructed the ω rows of H out of S_i with distinct coefficents, criterion 2 is satisfied. Criterion 3 is satisfied inductively for variables with exponents lower than $d-1$. Because $\min A_i = a_{2^i}$ for all i and $2i - 1 < 2^i$ for all naturals $i \geq 1$, $a_{2i-1} < a$ for all $a \in A_i$. So criterion 3 is satisfied by e. Because A_0 is disjoint from all A_i with $i \geq 1$, criterion 4 is satisfied for variables with exponent $d-1$, and by induction, it is satisfied for lower exponents. Because A_i is disjoint with A_j for all $i \neq j$, elements that differ in variables with exponent $d-1$ differ in all lower-exponent variables. The induction with the previous statement satisfies criterion 5. Therefore e satisfies a CR of $\binom{\omega^d}{n}$. The coefficients in e are contained in G by the construction of H from G. □

Theorem 8.3. *For $n, d \in \mathbb{N}$, $T\left(n, \omega^d\right) \leq P\left(n, \omega^d\right)$.*

Proof. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\omega^d}{n} \to [c]$$

be a c-coloring of $\binom{\omega^d}{n}$.

Enumerate the CRs of $\binom{\omega^d}{n}$ as τ_0 to $\tau_{P(n,\omega^d)-1}$. The maximum size of any CR of $\binom{\omega^d}{n}$ is $n \cdot d$. For each τ_i, let

$$f_i\colon \binom{\omega}{n \cdot d} \to \binom{\omega^d}{n}$$

where if τ_i has size p, f_i maps X to the unique $e \in \binom{\omega^d}{n}$ where e satisfies τ_i and the p equivalence classes of e are made up of the p least elements of X. For example, one CR of $\binom{\omega^2}{2}$ is

$$a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

The corresponding f_i would be $f_i\colon \binom{\omega}{4} \to \binom{\omega^2}{2}$ with

$$f_i(x_1, x_2, x_3, x_4) = \{\omega \cdot x_1 + x_2, \omega \cdot x_1 + x_3\}$$

where $x_1 < x_2 < x_3 < x_4$. Note that f_i does not depend on x_4 – this is because the example CR has size 3, but the largest CR for $\binom{\omega^2}{2}$ has size 4.

Then, define $\mathrm{COL}'\colon \binom{\omega}{n \cdot d} \to [c]^{P(n,\omega^d)}$ with

$$\mathrm{COL}'(X) = (\mathrm{COL}(f_0(X)), \mathrm{COL}(f_1(X)), \ldots, \mathrm{COL}(f_{P(n,\omega^d)-1}(X)))$$

and apply Theorem 2.6 to find some $G \approx \omega$ where

$$\left|\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)\right| = 1.$$

Let the one color in $\mathrm{COL}'\left(\binom{G}{n\cdot d}\right)$ be B. Note that B is a tuple of $P(n,\omega^d)$ colors.

Apply Lemma 8.2 to find some $H \approx \omega^d$ with the properties listed in Lemma 8.2. Now we claim

$$\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq P(n,\omega^d).$$

By Lemma 8.2, each element $e \in \binom{H}{n}$ satisfies a CR of $\binom{\omega^d}{n}$. Then for any arbitrary edge e, let e satisfy τ_i of size $p \leq n \cdot d$. Then take the p unique values in e, and if necessary, insert any new larger naturals from G to form a set of $n \cdot d$ values; denote this $A \in \binom{G}{n\cdot d}$. Then $\mathrm{COL}'(A) = B$ so by the definition of COL′, $\mathrm{COL}(e) \in B$. Because $|B| = P(n,\omega^d)$, $T(n,\omega^d) \leq P(n,\omega^d)$. □

8.2 $T(n,\omega^d) \geq P(n,\omega^d)$

Theorem 8.4. *For $n, d \in \mathbb{N}$, $T\left(n,\omega^d\right) \geq P\left(n,\omega^d\right)$. Therefore by Theorem 8.3, $T(n,\omega^d) = P(n,\omega^d)$.*

Proof. If $P(n,\omega^d) = 0$, then this is satisfied vacuously because $T(n,\omega^d) \geq 0$. Now suppose $P(n,\omega^d) \geq 1$. Note that all CRs of $\binom{\omega^d}{n}$ are disjoint from each other. That is, for any edge $e \in \binom{\omega^d}{n}$, if e satisfies τ', then it does not satisfy any nonequivalent CR of $\binom{\omega^d}{n}$, where equivalence between CRs is defined by their $\overline{B}$ maps and $\preceq_{\overline{A}}$ orders being the same. This is because if e were to satisfy two CRs τ_1 and τ_2, then τ_1 and τ_2 must share the same equivalence classes and order, so the CRs must be equivalent. Therefore, we can index them $\tau_0, \ldots, \tau_{P(n,\omega^d)-1}$ and construct a coloring $\mathrm{COL}\colon \binom{\omega^d}{n} \to [P(n,\omega^d)]$ with

$$\mathrm{COL}(e) = \begin{cases} i & e \text{ satisfies } \tau_i \\ 0 & \text{otherwise} \end{cases}$$

Similar to the proof of Theorem 6.2, our coloring has two ways to output color 0, both through the satisfaction of τ_0 and through the catch-all case. The part that forces color 0 to be present in all order-equivalent subsets is the satisfaction of τ_0.

We now prove there is no ω^d-$(P(n,\omega^d)-1)$-homogeneous set. For all $H \approx \omega^d$ and for every CR τ of $\binom{\omega^d}{n}$, we find some $e \in \binom{H}{n}$ that satisfies τ. For arbitrary $H \approx \omega^d$ and τ, we find $e_{i,j}$ where

$$\{\omega^{d-1}e_{1,d-1} + \cdots + \omega^1 e_{1,1} + e_{1,0}, \ldots, \omega^{d-1}e_{n,d-1} + \cdots + \omega^1 e_{n,1} + e_{n,0}\}$$

satisfies τ.

We do this by assigning values to each $e_{i,j}$ according to where the equivalence class that contains $a_{i,j}$ is found in τ, moving left to right in the permutation of τ. By criterion 3 of Definition 7.1, each $e_{i,j}$ is assigned before $e_{i,j-1}$. As we do this, we ensure that if the leftmost unassigned value in τ is $e_{i,j}$, then

$$\{\omega^{d-1}e_{i,d-1} + \cdots + \omega^{j+1}e_{i,j+1} + \omega^j c_j + \omega^{j-1}c_{j-1} + \cdots + \omega^1 c_1 + c_0 \ : \ c_i \in \omega\} \cap H \approx \omega^{j+1}.$$

By criterion 3 of Definition 7.1, the leftmost variable in τ must be $a_{i,d-1}$. Before any values are assigned, it is clear that

$$\{\omega^{d-1}c_{d-1} + \cdots + \omega^1 c_1 + c_0 : c_i \in \omega\} = \omega^d,$$

and because $H \subseteq \omega^d$, $\omega^d \cap H = H \approx \omega^d$.

By criterion 4 of Definition 7.1, all variables in an equivalence class must have the same exponent d. Let the leftmost unassigned equivalence class in τ be $a_{i_1,j} = a_{i_2,j} = \cdots = a_{i_m,j}$. By criterion 3, each $a_{i_i,\ell}$ for $1 \leq i \leq m$ and $\ell > j$ appeared to the left of this equivalence class and has already been assigned a value, and by criterion 5 the values for each exponent are equal: for all $\ell > j$ and $1 \leq i \leq m$, $e_{i_i,\ell} = e_{i_1,\ell}$.

By our previous steps,

$$\{\omega^{d-1}e_{i_1,d-1} + \cdots + \omega^{j+1}e_{i_1,j+1} + \omega^j c_j + \omega^{j-1}c_{j-1} + \cdots + \omega^1 c_1 + c_0 \ : \ c_i \in \omega\} \cap H \approx \omega^{j+1}.$$

Then there exists some value e' where

$$\{\omega^{d-1}e_{i_1,d-1} + \cdots + \omega^{j+1}e_{i_1,j+1} + \omega^j e' + \omega^{j-1}c_{j-1} + \cdots + \omega^1 c_1 + c_0 \ : \ c_i \ \in \ \omega\} \cap H \ \approx \ \omega^j,$$

where e' is greater than all previously assigned (and therefore finite) $e_{i,j}$ values. Then for $1 \leq i \leq m$, assign $e_{i_i,j}$ to be e'.

We can repeat this process to find $e_{i,j}$ for each CR of $\binom{\omega^d}{n}$ for arbitrary $H \approx \omega^d$. Therefore for all $H \approx \omega^d$, $\left|\text{COL}\left(\binom{H}{n}\right)\right| \geq P(n, \omega^d)$ so $T(n, \omega^d) \geq P\left(n, \omega^d\right)$. With Theorem 8.3, we have $T(n, \omega^d) = P\left(n, \omega^d\right)$.

□

9 Big Ramsey Degrees of $\omega^d \cdot k$

We now use the theory we developed for the case $k = 1$ to prove results for arbitrary k. We first extend the recurrence from Lemma 8.1.

Lemma 9.1. *For $n, d, k \in \mathbb{N}$,*

$$P_p\left(n, \omega^d \cdot k\right) =$$

$$\begin{cases} 0 & (1) \quad d = 0 \wedge n > k \\ 1 & (2) \quad n = 0 \wedge p = 0 \\ 0 & (3) \quad n = 0 \wedge p \geq 1 \\ \binom{k}{n} & (4) \quad d = 0 \wedge 1 \leq n \leq k \wedge p = 0 \\ 0 & (5) \quad d = 0 \wedge 1 \leq n \leq k \wedge p \geq 1 \\ k^n & (6) \quad d = 1 \wedge n \geq 1 \wedge n = p \\ 0 & (7) \quad d = 1 \wedge n \geq 1 \wedge n \neq p \\ 0 & (8) \quad d \geq 2 \wedge n \geq 1 \wedge p = 0 \\ k \sum\limits_{j=1}^{n} \sum\limits_{i=0}^{p-1} \binom{p-1}{i} P_i\left(j, \omega^{d-1}\right) P_{p-1-i}\left(n-j, \omega^d \cdot k\right) & (9) \quad d \geq 2 \wedge n \geq 1 \wedge p \geq 1 \end{cases}$$

Proof. **Case 1.** First, suppose $n > k$ and $d = 0$. Since $0 \leq b_i < k$ for all b_i, there are at most k unique values for the b_i. But by criterion 1 of Definition 7.1, since $d = 0$, we need n unique values of b_i, so no CRs are possible regardless of size p. This proves the first case of the result.

Case 2. Suppose $n = 0$. Then there can be no b_i, and since $n \cdot d = 0$, there can be no $a_{i,j}$. Thus all criteria are vacuously satisfied. Because there are no b_i or $a_{i,j}$, there is only one CR, and it has size $p = 0$. This proves the second and third cases of the result.

Case 3. When both $d = 0$ and $n \leq k$, criterion 1 of Definition 7.1 can be satisfied with the assignments to the n variables b_i being any permutation of n unique values out of k possible natural values. This leads to $\binom{k}{n}$ feasible combinations. Again, because $n \cdot d = 0$, there are no variables $a_{i,j}$ to permute so there are $\binom{k}{n}$ empty CRs of size $p = 0$, which proves the fourth and fifth cases of the result.

Case 4. Now suppose $n \geq 1$ and $d = 1$. To ensure criteria 1 of Definition 7.1, each of the n variables $a_{i,0}$ can only form one permutation $a_{1,0} < a_{2,0} < \ldots < a_{n,0}$ of size n. Because all $a_{i,j}$ are distinct and $d = 1$, the values b_i are not restricted by any criteria so each of the n variables can be any of the k naturals. Therefore $P_n\left(n, \omega^d \cdot k\right) = k^n$ and $P_p\left(n, \omega^d \cdot k\right) = 0$ for $p \neq n$. This proves the sixth and seventh cases of the result.

Case 5. Finally, consider $n \geq 1, d \geq 2$. By the definition of a CR , because $n \cdot d > 0$, there are no CRs of size $p = 0$. This proves the eighth case of our result. We prove the final case by showing the process for combining CRs described below creates all possible CRs of an expression.

Let $n \geq 1$, $d \geq 2$, $k \geq 0$ be naturals. Let $p \geq 1$, $1 \leq m \leq n$ and $0 \leq s \leq p-1$ be naturals. We create

$$k\binom{p-1}{s} P_s\left(m, \omega^{d-1}\right) P_{p-1-s}\left(n-m, \omega^d \cdot k\right)$$

CRs of size p by combining $P_s\left(m, \omega^{d-1}\right)$ CRs of size s and $P_{p-1-s}\left(n-m, \omega^d \cdot k\right)$ CRs of size $p-1-s$, with $k\binom{p-1}{s}$ new CRs for each pair of smaller CRs.

Let τ_1 represent one of the $P_s\left(m, \omega^{d-1}\right)$ CRs of $\binom{\omega^{d-1}}{m}$ of size s, and τ_2 represent one of the $P_{p-1-s}\left(n-m, \omega^d \cdot k\right)$ CRs of $\binom{\omega^d \cdot k}{n-m}$ of size $p-1-s$.

Then we can combine each τ_1 and τ_2 to form $k\binom{p-1}{s}$ unique new CRs of size p. First, we intertwine them as in the proof of Lemma 8.1: change the indices of the $a_{i,j}$ in τ_2 to $a_{i+m,j}$ and combine the equivalence classes of τ_1 and τ_2 while preserving their original orders. Insert a leading equivalence class $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$. This leads to $\binom{p-1}{s}$ new permutations of the $a_{i,j}$.

Because τ_1 was a CR for $\binom{\omega^{d-1}}{m}$, each of its b_i had a value of 0. Now that we are creating a CR for $\binom{\omega^d \cdot k}{n}$, we can choose the b_i to be composed of values between 0 and $k-1$. By criterion 2 of Definition 7.1, because all elements from τ_1 are bound together in a leading high-exponent equivalence class, the b_i must all be equal to one value. This leads to k options for that value; with the options of permuting the $a_{i,j}$, this makes $k\binom{p-s}{s}$ ways to create a new CR.

For the new CR's values for b_i, we assign each element originally from τ_2 with its original b value (likely at a different index due to changing the indices earlier). Then, the remaining elements from τ_1 are given all the same b value from one of the k options.

Each CR is unique by the τ_1 and τ_2 used to create it because the process is invertible: we can remove the leading equivalence class and separate the remaining variables into τ_1 and τ_2 by whether their indices were in the leading equivalence class and changing the indices. The b values for τ_2 can be found from the CR's b values after reversing the index change, and the b values for τ_1 are all 0.

We claim each CR created by this process has the properties described by Definition 7.1: Because the high-exponent equivalence class $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$ was added at the start of the CR, the high-exponent coefficients of each term are smaller than the low-exponent coefficients, where coefficients were defined in Definition 6.1. Criterion 1 is satisfied by changing the indices of the variables. Criterion 2 is met by assigning all elements from τ_1 the same b value. The remaining criteria are satisfied because τ_1 and τ_2 satisfied them and their internal orders were preserved in permuting the equivalence classes. Therefore this process does not overcount CRs.

We also claim that every CR of $\binom{\omega^d \cdot k}{n}$ is counted by this process: each can be mapped to some τ_1 and τ_2 that create it by a similar argument to proving that the process creates unique CRs 2 paragraphs above. Every CR of $\binom{\omega^d \cdot k}{n}$ must have a leading equivalence class of $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$ to satisfy Definition 7.1 (the equivalence class might only contain one variable); taking only the variables $a_{i,j}$ with indices appearing in that equivalence class (but not those variables in the equivalence class itself) with all-zero b values forms τ_1, a CR for $\binom{\omega^{d-1}}{m}$. The variables with i indices not in the equivalence class with their b values form τ_2, a CR for $\binom{\omega^d \cdot k}{n-m}$. The original CR of $\binom{\omega^d \cdot k}{n}$ was counted by interleaving τ_1 with τ_2 and inserting the leading equivalence class of $a_{i_1,d-1} = a_{i_2,d-1} = \cdots = a_{i_m,d-1}$. Therefore the final case of the result holds. □

9.1 $T(n, \omega^d \cdot k) \leq P(n, \omega^d \cdot k)$

Lemma 9.2. *For $n, d, k \in \mathbb{N}$ and some ordered subset $G \subseteq \omega$ with $G \approx \omega$, there exists some H with $H \subseteq \omega^d \cdot k$, $H \approx \omega^d \cdot k$ where for all $e \in \binom{H}{n}$, e satisfies a CR of $\binom{\omega^d \cdot k}{n}$ and each coefficient of e (see Definition 7.5) is contained in G.*

Proof. Let $G = \{a_0 < a_1 < a_2 < \cdots\}$. When $k = 0$, $H = \{\} = \emptyset$ suffices.

When $k > 0$, partition G into k distinct copies:

$$\begin{aligned} G_0 &= \{a_i : i \equiv 0 \mod k\} \\ G_1 &= \{a_i : i \equiv 1 \mod k\} \\ &\vdots \\ G_{k-1} &= \{a_i : i \equiv k-1 \mod k\}. \end{aligned}$$

Apply Lemma 8.2 on each G_j to yield $H_j \approx \omega^d$, and let

$$H = \{\omega^d \cdot j + h : h \in H_j\}.$$

We claim that all $e \in \binom{H}{n}$ satisfy a CR. Since we separated G into disjoint orders G_j, each H_j is disjoint from the others. With each $e' \in \binom{H_j}{n}$ satisfying a CR, criteria 1, 3, 4, and 5 of Definition 7.1 are automatically satisfied: if some criterion was not satisfiable and the H_j remained distinct, that would mean problematic elements came from the same part, contradicting how each $e' \in \binom{H_j}{n}$ satisfies a CR. Criterion 2 is satisfied by the construction of H and each of the H_j being distinct from each other. The coefficients of all edges e are contained in G by the construction of H from H_j, which have coefficients in G by Lemma 8.2.

□

Theorem 9.3. *For $n, d, k \in \mathbb{N}$, $T\left(n, \omega^d \cdot k\right) \leq P\left(n, \omega^d \cdot k\right)$.*

Proof. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\omega^d \cdot k}{n} \to [c]$$

be a c-coloring of $\binom{\omega^d \cdot k}{n}$.

Enumerate the CRs of $\binom{\omega^d \cdot k}{n}$ from τ_0 to $\tau_{P(n,\omega^d \cdot k)} - 1$. The maximum size of any CR of $\binom{\omega^d \cdot k}{n}$ is $n \cdot d$. For each τ_i, let its size be p_i. Define

$$f_i\colon \binom{\omega}{n \cdot d} \to \binom{\omega^d \cdot k}{n}$$

where f_i maps X to the unique $e \in \binom{\omega^d \cdot k}{n}$ where e satisfies τ_i and the p_i equivalence classes of e are made up of the p_i least elements of X. For example, one CR of $\binom{\omega^2 \cdot 2}{2}$ is

$$b_1 = 0,\ b_2 = 1,\ a_{1,1} < a_{2,1} < a_{2,0} < a_{1,0}.$$

The corresponding f_i would be $f_i\colon \binom{\omega}{4} \to \binom{\omega^2 \cdot 2}{2}$ with

$$f_i(x_1, x_2, x_3, x_4) = \{\omega^2 \cdot 0 + \omega \cdot x_1 + x_4, \omega^2 \cdot 1 + \omega \cdot x_2 + x_3\}$$

where $x_1 < x_2 < x_3 < x_4$. Note that the values of the b_i are used directly in the definition of f_i – for the CR with identical orderings on $a_{i,j}$ except $b_0 = 1$ and $b_1 = 0$, the coefficients b_i would be swapped.

Then, define $\mathrm{COL}'\colon \binom{\omega}{n \cdot d} \to [c]^{P(n,\omega^d \cdot k)}$ with

$$\mathrm{COL}'(X) = (\mathrm{COL}(f_0(X)), \mathrm{COL}(f_1(X)), \ldots, \mathrm{COL}(f_{P(n,\omega^d \cdot k)-1}(X)))$$

and apply Theorem 2.6 to find some $G \approx \omega$ where

$$\left|\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)\right| = 1.$$

Let B be the one color in $\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)$. Note that B is a tuple of $P(n, \omega^d \cdot k)$ colors.

Apply Lemma 9.2 to find some $H \approx \omega^d \cdot k$ with the properties listed in Lemma 9.2. Now we claim

$$\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq P(n, \omega^d \cdot k).$$

By Lemma 9.2, each element $e \in \binom{H}{n}$ satisfies a CR of $\binom{\omega^d \cdot k}{n}$. Then for any arbitrary edge e, let e satisfy τ_i of size $p_i \leq n \cdot d$. Then take the p_i unique values in e, and if necessary, insert any new larger nonnegative naturals from G to form a set of $n \cdot d$ values; denote this by $A \in \binom{G}{n \cdot d}$. Since $\mathrm{COL}'(A) = B$, by the definition of COL', $\mathrm{COL}(e) \in B$. Because $|B| = P(n, \omega^d \cdot k)$, $T(n, \omega^d \cdot k) \leq P(n, \omega^d \cdot k)$. □

9.2 $T(n, \omega^d \cdot k) \geq P(n, \omega^d \cdot k)$

Theorem 9.4. *For $n, d, k \in \mathbb{N}$, $T\left(n, \omega^d \cdot k\right) \geq P\left(n, \omega^d \cdot k\right)$. Therefore, by Theorem 9.3, $T(n, \omega^d \cdot k) = P(n, \omega^d \cdot k)$.*

Proof. If $P(n, \omega^d \cdot k) = 0$, then this is satisfied vacuously because $T(n, \omega^d \cdot k) \geq 0$. Suppose $P(n, \omega^d \cdot k) \geq 1$. Note that all CRs of $\binom{\omega^d \cdot k}{n}$ are disjoint from each other. That is, for any edge $e \in \binom{\omega^d \cdot k}{n}$, if e satisfies τ', then it does not satisfy any nonequivalent CR of $\binom{\omega^d \cdot k}{n}$. This is because if e were to satisfy two CRs τ_1 and τ_2, then τ_1 and τ_2 must share the same b_i values, equivalence classes, and order, so the CRs must be equivalent. Therefore, we can index them $\tau_0, \tau_1, \ldots, \tau_{P(n,\omega^d)-1}$ and construct a coloring COL: $\binom{\omega^d \cdot k}{n} \to [P(n, \omega^d \cdot k)]$ with

$$\mathrm{COL}(e) = \begin{cases} i & e \text{ satisfies } \tau_i \\ 0 & \text{otherwise.} \end{cases}$$

Similar to Theorem 6.2, our coloring has two ways to output color 0, both through the satisfaction of τ_0 and through the catch-all case. The part that forces color 0 to be present in all order-equivalent subsets is the satisfaction of τ_0. For arbitrary $H \approx \omega^d \cdot k$ and τ, we find variables b_i and $e_{i,j}$ such that

$$\{\omega^d b_1 + \omega^{d-1} e_{1,d-1} + \cdots + \omega^1 e_{1,1} + e_{1,0}, \ldots, \omega^d b_n + \omega^{d-1} e_{n,d-1} + \cdots + \omega e_{n,1} + e_{n,0}\}$$

satisfies τ.

For any $H \approx \omega^d \cdot k$ and τ, we first separate H into k ordered sets by the leading coefficient, each order-equivalent to ω^d.

Then, if there are equivalence classes in τ, using the process formally described in the proof of Theorem 8.4, we consider the leading equivalence class of τ. By criterion 2 of Definition 7.1, all variables in that equivalence class must come from the same set order-equivalent to ω^d. We assign a finite value to that equivalence class, and move to the next class with a potentially different b value using the assigned finite value as a lower bound for the next one. We can repeat this process to find $e_{i,j}$ that satisfy every CR of $\binom{\omega^d \cdot k}{n}$ for arbitrary $H \approx \omega^d$. Then, we can assign the b_i directly according to τ.

If there are no equivalence classes in τ (it has size $p = 0$), we can simply assign the variables b_i directly according to τ.

Therefore for all $H \approx \omega^d$, $\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \geq P(n, \omega^d \cdot k)$. Hence $T(n, \omega^d \cdot k) \geq P\left(n, \omega^d \cdot k\right)$. Therefore by Theorem 9.3, $T(n, \omega^d \cdot k) = P(n, \omega^d \cdot k)$. □

10 Big Ramsey Degrees of Ordinals less than ω^ω

10.1 General Coloring Rules

We defined CRs (coloring rules) to compute big Ramsey degrees of ordinals of the form $\omega^d \cdot k$. We now extend the definition to *GCRs* (general coloring rules), which allows us to compute big Ramsey degrees for all ordinals less than ω^ω.

Definition 10.1. We now define GCRs rigorously. Much like the definition of CRs, we impose a structure on edges, and then list criteria that GCRs must satisfy. Consider some ordinal $\alpha < \omega^\omega$:

$$\alpha \approx \omega^d \cdot k_d + \omega^{d-1} \cdot k_{d-1} + \cdots + \omega \cdot k_1 + k_0.$$

Then α is the addition of $d+1$ ordinals each of the form $\omega^i \cdot k_i$. For any element β of α, there is some i such that

$$\beta \in \omega^d \cdot k_d + \cdots + \omega^{i+1} \cdot k_{i+1} + \omega^i \cdot k_i \text{ and } \beta \notin \omega^d \cdot k_d + \cdots + \omega^{i+1} \cdot k_{i+1}.$$

In this case, we write that β *originated* from the $\omega^i \cdot k_i$ part of α. Note that the set of ordinals $\beta \in \alpha$ that originated from the $\omega^i \cdot k_i$ part of α is order-equivalent to $\omega^i \cdot k_i$.

For a natural number $n \geq 0$, there are n elements, denoted $p_i \in \alpha$, in any $e \in \binom{\alpha}{n}$. Unlike in the definition of CRs, we will need to allow each p_i to have anywhere from 0 to d variables $a_{i,j}$, depending on which of the

$d+1$ ordered sets the element originated from. We use $0 \le c_i \le d$ for the number of such variables that the element p_i has. Elements $p_i \in \alpha$ with c_i variables originated from the $\omega^{c_i} \cdot k_{c_i}$ part of α. We denote each element of e as

$$\omega^d \cdot k_d + \cdots + \omega^{c_i+1} \cdot k_{c_i+1} + \omega^{c_i} \cdot b_i + \omega^{c_i-1} \cdot a_{i,c_i-1} + \omega^{c_i-2} \cdot a_{i,c_i-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0}.$$

where $0 \le b_i < k_{c_i}$ and $0 \le a_{i,j}$. Note that for all elements originating from the $\omega^{c_i} \cdot k_{c_i}$ part of α, the leading terms

$$\omega^d \cdot k_d + \cdots + \omega^{c_i+1} \cdot k_{c_i+1}$$

are always the same; only the variables c_i, b_i, and $a_{i,j}$ are needed to uniquely identify an element.

A *general coloring rule*, hereafter referred to as a GCR, is a triple $(\overline{C}, \overline{B}, \preceq_{\overline{A}})$ of constraints on the c_i, b_i, and $a_{i,j}$.

The function $\overline{C} : [n] \to \{0, \ldots, d\}$ maps indices i to their corresponding values c_i. The function $\overline{B}$ maps $\overline{B} : [n] \to \omega$ such that $\overline{B}(i) < k_{\overline{C}(i)}$, mapping from indices i to the values b_i. The order $\preceq_{\overline{A}}$ is a total preorder on the indices of the $a_{i,j}$. We continue to use the same notation to represent the preorder.

GCRs must fulfill the following criteria (only criterion 6 below is different from its corresponding criterion in Definition 7.1, the definition for CRs):

1. If $d \ge 1$, then for all $i < i'$, $a_{i,0} < a_{i',0}$. Otherwise when $d = 0$, $b_i < b_{i'}$ for all $i < i'$. (The element indices are ordered by their lowest-exponent variable.)
2. $a_{i,j} = a_{i',j} \implies b_i = b_{i'}$ for all j. (Elements with any a values that are the same must have the same b value.)
3. For all $j > j'$, $a_{i,j} < a_{i,j'}$. (The high-exponent variables of each element are strictly less than the low-exponent variables.)
4. $a_{i,j} = a_{i',j'} \implies j = j'$. (Only variables with the same exponent can be equal.)
5. $a_{i,j} \ne a_{i',j} \implies a_{i,j-1} \ne a_{i',j-1}$ for all $j > 0$. (Elements that differ in a high-exponent variable differ in all lower-exponent variables.)
6. $c_i \ne c_{i'} \implies (b_i \ne b_{i'}$ and $a_{i,j} \ne a_{i',j})$ for all $0 \le j < d$. (Different c variables mean different b and a variables.)

Definition 10.2. Two GCRs are *equivalent* whenever their constraints $\overline{C}, \overline{B}, \preceq_{\overline{A}}$ are exactly equal.

Definition 10.3. We again define the *size* of a GCR to be how many equivalence classes its a variables form. A GCR's size p is still bounded above by $d \cdot n$.

Definition 10.4.

1. $S_p(n, \alpha)$ is the number of GCRs of size p there are for $\binom{\alpha}{n}$.
2. $S(n, \alpha)$ is the total number of GCRs there are for $\binom{\alpha}{n}$ regardless of size. It can be calculated as

$$\sum_{p=0}^{n \cdot d} S_p(n, \alpha).$$

We will show $T(n, \alpha) = S(n, \alpha)$.

Definition 10.5. For an edge

$$e = \{\omega^{c_i} \cdot b_i + \omega^{c_i-1} \cdot a_{i,c_i-1} + \omega^{c_i-2} \cdot a_{i,c_i-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0} \colon 1 \le i \le n\},$$

e *satisfies* the GCR $(\overline{C}, \overline{B}, \preceq_{\overline{A}})$ if $c_i = \overline{C}(i)$ and $b_i = \overline{B}(i)$ for every $1 \le i \le n$, and if $(i, j) \preceq_{\overline{A}} (i', j') \iff a_{i,j} \le a_{i',j'}$ for all $1 \le i, i' \le n$, $0 \le j < c_i$, and $0 \le j' < c_{i'}$. Again, the coefficients of an edge satisfying some GCR is the set of its $a_{i,j}$ terms.

Lemma 10.6. *For $n, d, k, p \in \mathbb{N}$, $S_p(n, \omega^d \cdot k) = P_p(n, \omega^d \cdot k)$.*

Proof. Let $\alpha = \omega^d \cdot k$. Suppose some $c_i \neq d$. Then $b_i < k_{c_i}$ by Definition 10.1, so $b_i < 0$, which is impossible. Thus $c_i = d$. Then there are the same count of $n \cdot d$ variables $a_{i,j}$ being permuted, the new criterion 6 has no effect because all c_i are equal. Hence both are under the same restrictions so $S_p(n, \omega^d \cdot k) = P_p(n, \omega^d \cdot k)$. □

Lemma 10.7. *For all $\alpha < \omega^\omega$ with*

$$\alpha \approx \omega^d \cdot k_d + \omega^{d-1} \cdot k_{d-1} + \cdots + \omega \cdot k_1 + k_0 \text{ with } k_d \neq 0 \text{ and } d > 0,$$

$$S_p(n, \alpha) = \sum_{j=0}^{n} \sum_{i=0}^{p} \binom{p}{i} P_i\left(j, \omega^d \cdot k_d\right) S_{p-i}\left(n-j, \omega^{d-1} \cdot k_{d-1} + \cdots + \omega \cdot k_1 + k_0\right).$$

When the conditions $k_d \neq 0$ and $d > 0$ cannot be satisfied, then $\alpha < \omega$ and $S_p(n, k_0) = P_p(n, k_0)$.

Proof. When k_0 is the only nonzero k term, Lemma 10.6 shows $S_p(n, \alpha) = P_p(n, \alpha)$. When $d \geq 1$, we describe a process of combining CRs with GCRs to create GCRs for $\binom{\alpha}{n}$.

Let $n \geq 0, p \geq 0$ be naturals. For some $\alpha < \omega^\omega$, let $0 \leq m \leq n$ and $0 \leq s \leq p$ be naturals. We create

$$\binom{p}{s} P_s\left(m, \omega^d \cdot k_d\right) S_{p-s}\left(n-m, \omega^{d-1} \cdot k_{d-1} + \cdots + k_0\right)$$

GCRs, with each GCR having m elements from the $\omega^d \cdot k_d$ part of α and $n - m$ elements from parts with lower exponents.

Let τ_1 represent one of the $P_s\left(m, \omega^d \cdot k_d\right)$ CRs of $\binom{\omega^d \cdot k_d}{m}$ of size s, and τ_2 represent one of the $S_{p-s}\left(n-m, \omega^{d-1} \cdot k_{d-1} + \cdots\right.$ GCRs of $\binom{\omega^{d-1} \cdot k_{d-1} + \cdots + k_0}{n-m}$ of size $p - s$. We change τ_1 into a GCR by assigning it $c_i = d$ for all c_i.

Then we can combine each τ_1 and τ_2 to form $\binom{p}{s}$ unique new GCRs of size p: Change the indices τ_2 and permute the equivalence classes as in the proof of Lemma 8.1. Note that we do not insert a leading equivalence class – this is because we do not need to increase the exponent or size of τ_1.

We can keep each c_i and b_i value the same, and change their indices alongside the $a_{i,j}$ variables to ensure criterion 1.

Each GCR is unique by the τ_1 and τ_2 used to create it because the process is invertible: we can identify the elements originally from τ_1 because they uniquely have $c_i = d$.

We claim each GCR created by this process has the properties described by Definition 10.1: Because all c_i are equal for τ_1, criterion 6 is satisfied for the elements from τ_1. Criterion 1 is satisfied by changing the indices of the variables. The remaining criteria are satisfied because τ_1 and τ_2 satisfied them and their internal orders and equivalence classes were preserved in permuting the equivalence classes. Therefore this process does not overcount GCRs.

We also claim that every GCR of $\binom{\alpha}{n}$ is counted by this process: each can be mapped to some τ_1 and τ_2 that create it by a similar argument to proving that the process creates unique GCRs 2 paragraphs above. □

10.2 $T(n, \alpha) \leq S(n, \alpha)$

Lemma 10.8. *For all $\alpha < \omega^\omega$, $n \in \mathbb{N}$, and ordered subsets $G \subseteq \omega$ with $G \approx \omega$, there exists some $H \subseteq \alpha$, $H \approx \alpha$ where for all $e \in \binom{H}{n}$, e satisfies a GCR of $\binom{\alpha}{n}$ and each coefficient of e (see Definition 10.5) is contained in G.*

Proof. Because $G \approx \omega$, $G = \{a_0 < a_1 < a_2 < \cdots\}$. Let $\alpha \approx \omega^d \cdot k_d + \omega^{d-1} \cdot k_{d-1} + \cdots + \omega \cdot k_1 + k_0$.

Partition G into $d + 1$ distinct copies:

$$\begin{aligned} G_0 &= \{a_i : i \equiv 0 \mod d+1\} \\ G_1 &= \{a_i : i \equiv 1 \mod d+1\} \\ &\vdots \\ G_d &= \{a_i : i \equiv d \mod d+1\}. \end{aligned}$$

For $0 \leq j \leq d$, apply Lemma 9.2 on G_j to yield some $H_j \approx \omega^j \cdot k_j$ where all $e \in \binom{H_j}{n}$ satisfy a CR for $\binom{\omega^j \cdot k_j}{n}$. We then sum the remaining portion of α with each H_j, aligning with the order of α:

$$H'_j = \{\omega^d \cdot k_d + \cdots + \omega^{j+1} \cdot k_{j+1} + \beta : \beta \in H_j\} \approx H_j.$$

Then let

$$H = H'_d \cup H'_{d-1} \cup \cdots \cup H'_0,$$

ordered by ordinal comparison. Since every element of H'_{i+1} is less than every element of H'_i, $H \approx \alpha$.

We claim that all $e \in \binom{H}{n}$ satisfy a GCR. Since we separated G into disjoint orders G_j, each H'_j is disjoint from the others. With each $e' \in \binom{H_j}{n}$ satisfying a CRof a lesser ordinal by Lemma 9.2, this continues to each $e' \in \binom{H'_j}{n}$ with the same $b_i, a_{i,j}$. Hence Criteria 1, 2, 3, 4, and 5 of Definition 10.1 are automatically satisfied for e: if some criterion was not satisfied and the H'_j remained distinct, that would mean problematic elements came from the same part, contradicting how each $e' \in \binom{H'_j}{n}$ satisfies a CR. Criterion 6 is satisfied by each of the H'_j being distinct from each other. The coefficients of all edges e are contained in G by the construction of H from G. □

Theorem 10.9. *For all $\alpha < \omega^\omega$, $T(n, \alpha) \leq S(n, \alpha)$.*

Proof. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\alpha}{n} \to [c]$$

be a c-coloring of $\binom{\alpha}{n}$.

Enumerate the GCRs of $\binom{\alpha}{n}$ from τ_0 to $\tau_{S(n,\alpha)-1}$. The maximum size of any GCR of $\binom{\alpha}{n}$ is $n \cdot d$. For each τ_i, let

$$f_i\colon \binom{\omega}{n \cdot d} \to \binom{\alpha}{n}$$

where if τ_i has size p, f_i maps X to the unique $e \in \binom{\alpha}{n}$ where e satisfies τ_i and the p equivalence classes of e are made up of the p least elements of X. For example, one GCR of $\binom{\omega^2+\omega\cdot 8}{2}$ is

$$c_1 = 2, c_2 = 1, b_1 = 0, b_2 = 6,\ a_{1,1} < a_{2,0} < a_{1,0}.$$

The corresponding f_i would be $f_i\colon \binom{\omega}{4} \to \binom{\omega^2+\omega\cdot 8}{2}$ with

$$f_i(x_1, x_2, x_3, x_4) = \{\omega^2 \cdot 0 + \omega \cdot x_1 + x_3, \omega^2 + \omega \cdot 6 + x_2\}$$

where $x_1 < x_2 < x_3 < x_4$.

Then, define $\mathrm{COL}'\colon \binom{\omega}{n\cdot d} \to [c]^{S(n,\alpha)}$ with

$$X = (\mathrm{COL}(f_0(X)), \mathrm{COL}(f_1(X)), \ldots, \mathrm{COL}(f_{S(n,\alpha)-1}(X)))$$

and apply Theorem 2.6 to find some $G \approx \omega$ where

$$\left|\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)\right| = 1.$$

Let B be the one color in $\mathrm{COL}'\left(\binom{G}{n\cdot d}\right)$. Note that B is a tuple of $S(n, \alpha)$ colors.

Apply Lemma 10.8 to find some $H \approx \alpha$ with the properties listed in Lemma 10.8. Now we claim

$$\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq S(n, \alpha).$$

By Lemma 10.8, each element $e \in \binom{H}{n}$ satisfies a GCR of $\binom{\alpha}{n}$. Then for any edge e, let e satisfy τ_i of size $p \leq n \cdot d$. Then take the p unique values in e, and if necessary, insert any new larger nonnegative naturals from G to form a set of $n \cdot d$ values; denote this $A \in \binom{G}{n\cdot d}$. Then $\mathrm{COL}'(A) = B$ so by the definition of COL', $\mathrm{COL}(e) \in B$. Because $|B| = S(n, \alpha)$, $T(n, \alpha) \leq S(n, \alpha)$. □

10.3 $T(n,\alpha) \geq S(n,\alpha)$

Theorem 10.10. *For all $\alpha < \omega^\omega, T(n,\alpha) \geq S(n,\alpha)$, and thus by Theorem 10.9, $T(n,\alpha) = S(n,\alpha)$.*

Proof. If $S(n,\alpha) = 0$, this is satisfied vacuously because $T(n,\alpha) \geq 0$. Suppose $S(n,\alpha) \geq 1$. Note that all GCRs of $\binom{\alpha}{n}$ are disjoint from each other. That is, for any edge $e \in \binom{\alpha}{n}$, if e satisfies τ', then it does not satisfy any nonequivalent GCR of $\binom{\alpha}{n}$. This is because if e were to satisfy two GCR τ_1 and τ_2, then τ_1 and τ_2 must share the same $\overline{C}, \overline{B}$ functions and order $\preceq_{\overline{A}}$, so the GCRs must be equivalent. Therefore, we can index them $\tau_0, \ldots, \tau_{S(n,\alpha)-1}$ and construct a coloring $\mathrm{COL}\colon \binom{\alpha}{n} \to [S(n,\alpha)]$ with

$$\mathrm{COL}(e) = \begin{cases} i & e \text{ satisfies } \tau_i \\ 0 & \text{otherwise} \end{cases}$$

For arbitrary $H \approx \alpha$ and a GCR τ for α, we can assign c_i and b_i based on τ. Then we can apply a similar process to the one used in Theorem 9.4 to find $e_{i,j}$ variables that match the permutation of $a_{i,j}$ variables.

Let $\alpha \approx \omega^d \cdot k_d + \omega^{d-1} \cdot k_{d-1} + \cdots + \omega \cdot k_1 + k_0$. We can separate H into $d+1$ sets each order-equivalent to $\omega^j \cdot k_j$ for $0 \leq j \leq d$, and separate each of those into k_j sets order-equivalent to ω^j.

Then, for each equivalence class in τ, using the process formally described in the proof of Theorem 8.4, we consider the leading equivalence class of τ. By criteria 2 and 6 of Definition 10.1, all variables in that equivalence class must come from the same set order-equivalent to ω^j. We assign a finite value to that equivalence class and move to the next class with potentially different c and b values, using the assigned finite value as a lower bound for the next one. We can repeat this process to find $e_{i,j}$ that satisfy each GCR of $\binom{\alpha}{n}$ for arbitrary $H \approx \alpha$.

Therefore for all $H \approx \alpha$, $\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \geq S(n,\alpha)$ so $T(n,\alpha) \geq S(n,\alpha)$. With Theorem 10.9, we have $T(n,\alpha) = S(n,\alpha)$. □

Appendix

Table 1 shows $T(n,\omega^d)$ for small n, d.

$T(n,\omega^d)$		n					
		0	1	2	3	4	5
d	0	1	1	0	0	0	0
	1	1	1	1	1	1	1
	2	1	1	4	26	236	2752
	3	1	1	14	509	35839	4154652
	4	1	1	49	10340	5941404	7244337796
	5	1	1	175	222244	1081112575	14372713082763

Table 1: Big Ramsey degrees of ω^d.

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